

Part 1: Advanced Python Data Structures

Task 1: Lists and Tuples

1. Create a list of integers from 1 to 100. Slice the list to get every 5th element and print it.
2. Convert the list to a tuple and print the first and last elements.
3. Create a nested list comprehension to generate a 3x3 matrix with values from 1 to 9.

Task 2: Dictionaries

1. Create a dictionary with student names as keys and their scores as values. The dictionary should have at least 5 entries.
2. Calculate the average score and print it.
3. Find and print the student(s) with the highest score.

Part 2: Advanced NumPy Usage

Task 1: Creating Complex NumPy Arrays

1. Import the NumPy library.
2. Create a 4x4 matrix with random integers between 1 and 100.
3. Create a 4x4 identity matrix and add it to the random matrix. Print the resulting matrix.

Task 2: Array Slicing and Indexing

1. Slice the matrix to get the second and third rows and columns, and print the submatrix.
2. Replace all even numbers in the matrix with -1 and print the updated matrix.

Task 3: Broadcasting and Vectorized Operations

1. Create two 1D arrays, one with 10 random integers and another with integers from 10 to 1.
2. Perform element-wise addition, subtraction, multiplication, and division on the arrays and print the results.
3. Use broadcasting to add a 1D array to each row of a 2D array and print the result.

Part 3: Advanced Pandas Usage

Task 1: Complex DataFrame Manipulations

1. Import the Pandas library.
2. Create a DataFrame with columns 'Name', 'Age', 'Department', 'Salary', and 'Years of Service' with at least 7 entries.
3. Add a new column 'Bonus' which is 10% of the 'Salary' if 'Years of Service' is greater than 5, otherwise 5%.

Task 2: Handling Missing Data

1. Introduce NaN values randomly into the DataFrame.

2. Fill NaN values in 'Salary' with the mean salary and drop rows where 'Department' is NaN.
3. Print the cleaned DataFrame.

Task 3: Grouping and Aggregation

1. Group the DataFrame by 'Department' and calculate the average 'Salary' and total 'Years of Service' for each department.
2. Print the grouped and aggregated data.

Part 4: Advanced Data Visualization with Matplotlib

Task 1: Creating Complex Plots

1. Import the Matplotlib library.
2. Create a scatter plot with random points. Color the points based on their distance from the origin.
3. Add a regression line to the scatter plot.

Task 2: Customizing and Annotating Plots

1. Create a bar chart showing the average salary for each department in the DataFrame from Part 3.
2. Customize the bar chart with different colors, add a legend, and annotate each bar with the exact value.

Task 3: Creating Interactive Visualizations

1. Use Matplotlib's interactive features to create a plot where you can toggle the visibility of different data series.
2. Implement interactivity such that clicking on a point in the scatter plot displays its coordinates.

Submission:

Submit the completed Jupyter notebook containing all the tasks and their outputs. Ensure your code is well-commented and follows best practices.